

Books and topics to be covered

- Artificial Intelligence by Russel and Norvig
- Artificial Intelligence by Luger
- Broad topics
 - Types of intelligent agents
 - General problem solving
 - Logic and reasoning
 - Planning
 - Learning : machine learning, deep learning
 - AI languages (LISP, Prolog, CLIPS)
 - Communication (NLP)
 - Multiple agents, web agents, negotiating agents

Artificial Intelligence

- Modeling human cognition or mental faculty using computers
- Study of making computers do things at which at the moment people are better
- Making computers do things which require intelligence

More Formal Definition of AI

- AI is a branch of computer science which is concerned with the study and creation of computer systems that exhibit
 - some form of intelligence
- OR
- those characteristics which we associate with intelligence in human behavior

Definition

AI is a broad area consisting of different fields, from machine vision, expert systems to the creation of machines that can "think".

In order to classify machines as "thinking", it is necessary to define intelligence.

Artificial Intelligence

- **Behavior** by a machine that, if performed by a human being, would be considered **intelligent**
- “...study of how to make computers do things at which, at the moment, people are better”
(Rich and Knight [1991])
- Theory of how the **human mind** works (Mark Fox)

Definitions (contd.)

- Actions that are indistinguishable from a human's
–Alan Turing (Cambridge, 1950)

- “AI is the study of complex information processing problems that often have their roots in some aspect of biological information processing. The goal of the subject is to identify solvable and interesting information processing problems and solve them.”
–David Marr (MIT, 1977)

- The intelligent connection of perception to action
–Rodney Brooks (MIT, 1990)

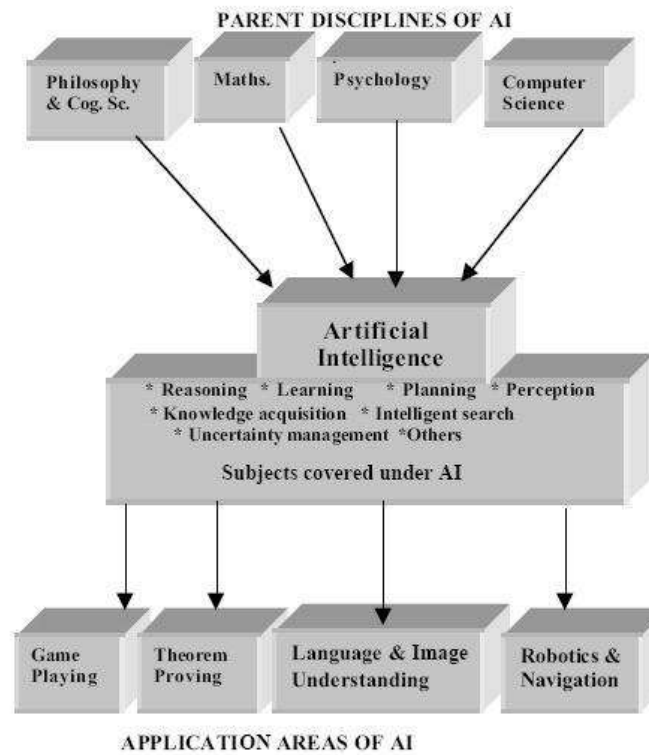
Characteristics of AI systems

- learn new concepts and tasks
- reason and draw useful conclusions about the world around us
- remember complicated interrelated facts and draw conclusions from them (inference)
- understand a natural language or perceive and comprehend a visual scene
- look through cameras and see what is there (vision), to move themselves and objects around in the real world (robotics)

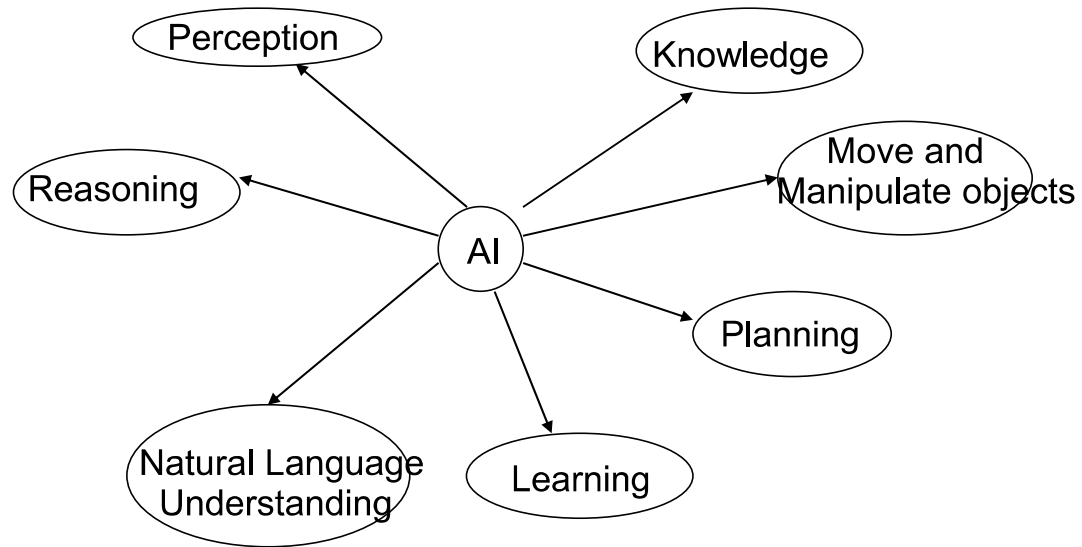
Contd..

- plan sequences of actions to complete a goal
- offer advice based on rules and situations
- may not necessarily imitate human senses and thought processes but indeed, in performing some tasks differently, they may actually exceed human abilities
- capable of performing intelligent tasks effectively and efficiently
- perform tasks that require high levels of intelligence

Overview of AI



Sub-areas in AI



Sub-areas of AI

- Knowledge representation
- Theorem proving
- Game playing
- Common sense reasoning dealing with uncertainty and decision making
- Learning models, inference techniques, pattern recognition, search and matching etc.
- Logic (fuzzy, temporal, modal) in AI
- Planning and scheduling

Sub-areas of AI

- Natural language understanding
- Computer vision
- Understanding spoken utterances
- Intelligent tutoring systems
- Robotics
- Machine translation systems
- Expert problem solving
- Neural Networks, AI tools etc

What is Intelligence? (Dictionary)

- Intelligence: (Webster's Dictionary)
 1. the ability to learn or understand or to deal with new or trying situations : reason also : the skilled use of reason
 2. the ability to apply knowledge to manipulate one's environment or to think abstractly as measured by objective criteria (such as tests)
 3. The capacity to learn and solve problems
 4. The ability to act rationally

What is Intelligence?

- **Intelligence:**
 - “the capacity to learn and solve problems” (Websters dictionary)
 - in particular,
 - *the ability to solve novel problems*
 - *the ability to act rationally*
 - *the ability to act like humans*
- **Artificial Intelligence**
 - build and understand intelligent entities or agents
 - 2 main approaches: “engineering” versus “cognitive modeling”

What's involved in Intelligence

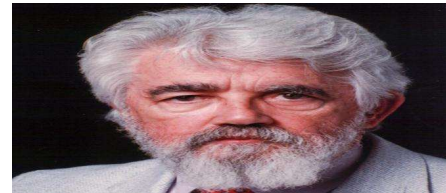
- Ability to interact with the real world
- to perceive, understand, and act
 - e.g., speech recognition and understanding and synthesis
 - e.g., image understanding
 - e.g., ability to take actions, have an effect
- Reasoning and Planning
 - modeling the external world, given input
 - solving new problems, planning, and making decisions
 - ability to deal with unexpected problems, uncertainties
- Learning and Adaptation
 - we are continuously learning and adapting
 - our internal models are always being “updated”
 - e.g., a baby learning to categorize and recognize animals

Pioneers of AI

John McCarthy (MIT and Stanford)

(September 4, 1927 – October 24, 2011)

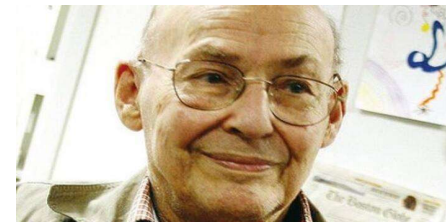
- LISP (Garbage Collection)
- idea of using logical reasoning to decide on actions
- Non-Monotonic Reasoning(Circumscription)



Marvin Minsky (MIT)

(August 9, 1927 – January 24, 2016)

- Co-Founder of AI Lab at MIT (CSAIL)
- Co-authored Perceptrons book which analyses artificial neural networks
- FRAME as a knowledge structure



Herbert Simon (CMU)

(June 15, 1916 – February 9, 2001)

- Logic Machines and General Problem Solver
- Preferential Attachment that can explain power law degree distribution of social nets



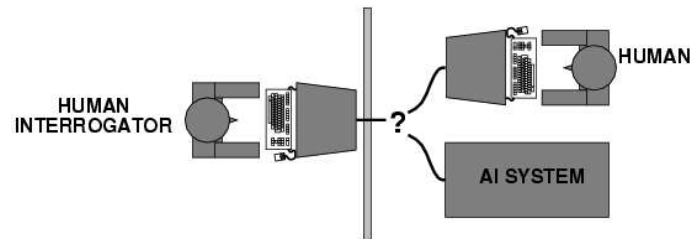
Pioneers of AI

- **Arthur Samuel (MIT, IBM, Stanford)**
- (December 5, 1901 – July 29, 1990)
- Coined the term Machine Learning in 1959
- AI for Game Playing
- Rote Learning
- **Allen Newell (CMU)**
- (March 19, 1927 – July 19, 1992)
- Logic Theorist (Logic based Reasoning)
- GPS: Knowledge and Reasoning
- Chess Machine
- **Nils Nilsson (Stanford)**
- (6 February 1933- 23 April 2019)
- A* Heuristic Search Algorithm
- Planning and Robotics (STRIPS)
- Early influential books on AI and Learning



Turing test

- Operational test for intelligent behavior: the Imitation Game

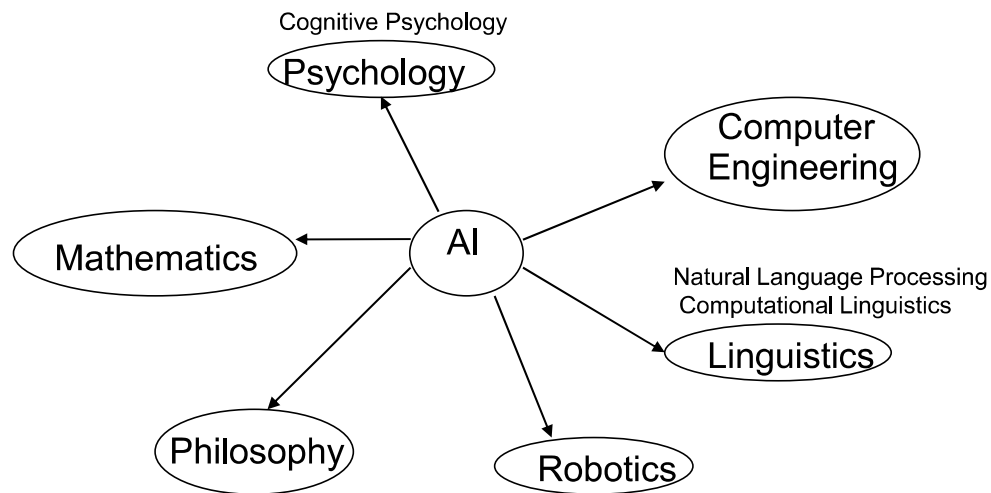


- Computer interrogated by a human via a teletype
 - If interrogator cannot tell if there is a computer or human at the other end, computer passes the test
- * Question: is it important that an intelligent system act like a human?

Turing Test

- Capabilities required by computer
 - **Natural language processing**: to communicate in English with interrogator
 - **Knowledge representation**: to store information obtained before or during interrogation
 - **Automated reasoning**: to use stored information to answer questions and draw conclusions
 - **Machine learning**: to adapt to new circumstances

AI : multidisciplinary field



- AI exploits synergy between
 - philosophy, psychology, cognitive science
 - computer science, data science, machine learning, deep learning
 - Mathematics, physics
 - Linguistics, robotics, etc

Academic Disciplines relevant to AI

- **Philosophy** Logic, methods of reasoning, mind as physical system, foundations of learning, language, rationality.
- **Mathematics** Formal representation and proof, algorithms, computation, (un)decidability, (in)tractability
- **Probability/Statistics** modeling uncertainty, learning from data
- **Economics** utility, decision theory, rational economic agents
- **Neuroscience** neurons as information processing units.
- **Psychology/
Cognitive Science** how do people behave, perceive, process cognitive information, represent knowledge.
- **Computer engineering** building fast computers
- **Control theory** design systems that maximize an objective function over time
- **Linguistics** knowledge representation, grammars

Foundations of AI

- Foundation of AI is based on
 - Mathematics
 - Neuroscience
 - Control Theory
 - Linguistics

Foundations - Mathematics

- More formal logical methods
 - Boolean logic
 - Fuzzy logic
- Uncertainty
 - The basis for most modern approaches to handle uncertainty in AI applications can be handled by
 - ✓ Probability theory
 - ✓ Modal and Temporal logics

Foundations - Neuroscience

- How does the brain works
 - Early studies (1824) relied on injured and abnormal people to understand what parts of brain work
 - More recent studies use accurate sensors to correlate brain activity to human thought
 - By monitoring individual neurons, monkeys can now control a computer mouse using thought alone
 - How close are we to have a mechanical brain?
 - Parallel computation, remapping, interconnections,....

Foundations – Control Theory

- Machines can modify their behavior in response to the environment (sense/action loop)
 - Water-flow regulator, steam engine governor, thermostat
- The theory of stable feedback systems (1894)
 - Build systems that transition from initial state to goal state with minimum energy
 - In 1950, control theory could only describe linear systems and AI largely rose as a response to this shortcoming

Foundations - Linguistics

- Speech demonstrates so much of human intelligence
- Analysis of human language reveals thought taking place in ways not understood in other settings
 - Children can create sentences they have never heard before
 - Language and thought are believed to be tightly intertwined